

MIS 632 – Database Analysis and Design for Business

Project

1. Overview:

For this project, students will work in groups of four. The objective of this group project is to select a business operation task, and conduct an analysis of this operation in order to design and develop a relational database to support the chosen business operation.

The following are examples of business operation tasks for this database analysis and design project:

1. Fitness gym management
2. Chain restaurant online ordering
3. Car rental service
4. Internet TV subscription
5. NBA player performances
6. Olympics swimming meetings
7. Movie theater ticketing
8. Public library book check-out/return service
9. Apartment rental or leasing management

Each group can select one business operation task to design and develop a relational database to support this operation. The business task can be from the above list. Discuss with the instructor if you would consider a business operation that is not on this list.

For the selected business operation, you will analyze the requirements for the supporting database, design the database models, create the database in Oracle APEX, and identify business questions that can be studied with the data in the database. This project has two deliverables: one project report, and one completed database. The detailed requirements are given in the section “Evaluation”.

2. Project Timeline:

You may form groups on your own, and choose the business operation task for database design as a team.

By **Monday, April 27th**, each group is expected to submit your group member names and selected business operation task for the project.

On **Monday, May 18th**, in class, each group will discuss with the instructor the project progress and continue towards completion with any feedback.

By **Monday, June 8th**, the project deliverables are due. Each group is required to submit a project report and a completed database in Oracle APEX.

3. Evaluation

3.1 Deliverables:

This project has two deliverables: one project report, and one completed database.

The project report should include the following components:

1. Description of the business operation task for the database design. Analyze the requirements for the database design. (10 points)
2. According to the above requirements, design the entity-relationship model. The ER model should include four to six entities. (20 points)
3. Given the ER model, design the relational schema. Provide descriptions about tables, fields (columns) including data types and constraints, primary keys, foreign keys, indexes. (10 points)
4. Identify three business questions that can be answered with the data in the database; Write SQL statements to retrieve the data to answer these questions. At least one SQL statement should use data from multiple tables. At least one SQL statement should use GROUP BY clause. (10 points)

The database implements the design and should include all database objects from the data model that is developed for the chosen business operation as described in the report:

5. Tables, columns, primary keys, foreign keys, indexes. These objects should be aligned to the design in the report as specified in point #2-3. (30 points)
6. Populate the database with data. Each table contains at least five records. (10 points)
7. The data support the execution of the SQL statements given in point #4. (10 points)

3.2. Group peer evaluation

This is a group project and emphasizes teamwork. Your grade on the group project will be partly the result of your individual contribution and partly the result of your teammates' performance. While the grade of the group project applies to all members in the same group, we use confidential group peer evaluation to adjust the group project grade for individual members. The total team effort encompasses (not limited to) any of the following that might apply: leadership; arranging meetings; searching for materials; designing the data models; creating the database; collecting data; writing; and helping others learn. Each group member is expected to carry a fair share of the workload.

The next page is the group peer evaluation form that you may fill in at the end of the term. If you do not fill in the form and turn it in at the end of the term, it is assumed that you choose to allocate the points equally among the other group members.

MIS632 Confidential Group Peer Evaluation

To be reviewed only by the instructor

Group Name: _____ Your Name: _____ Date: _____

Each group member may rate all other members of the group, considering their individual contributions to the total team effort. The TOTAL TEAM EFFORT encompasses any of the following that might apply: leadership; arranging meetings; searching for materials; designing the data models; creating the database; collecting data; writing; and helping others learn. Each group member is expected to carry a fair share of the workload.

Course requirements: This group project emphasizes teamwork. Your grade on the group project will be partly the result of your individual contribution and partly the result of your teammates' performance. While the grade of the group project applies to all members in the same group, we use confidential group peer evaluation to adjust the group project grade for individual members.

Rating relative to other group members. Assess the contribution of each person relative to the total team effort. Allocate a total of 100 points among all group members other than yourself. Remember that group project constitutes 20% of the course grade.

If you do not fill in the form and turn it in at the end of the term, it is assumed that you choose to allocate the points equally among the other group members.

Carefully assign values to reflect the differences among group members.
Write down the member names and relative contributions clearly.

Other Group Member Names	Relative Contribution

Remark: Add explanatory remarks below if you rated a group member significantly differently than other members.